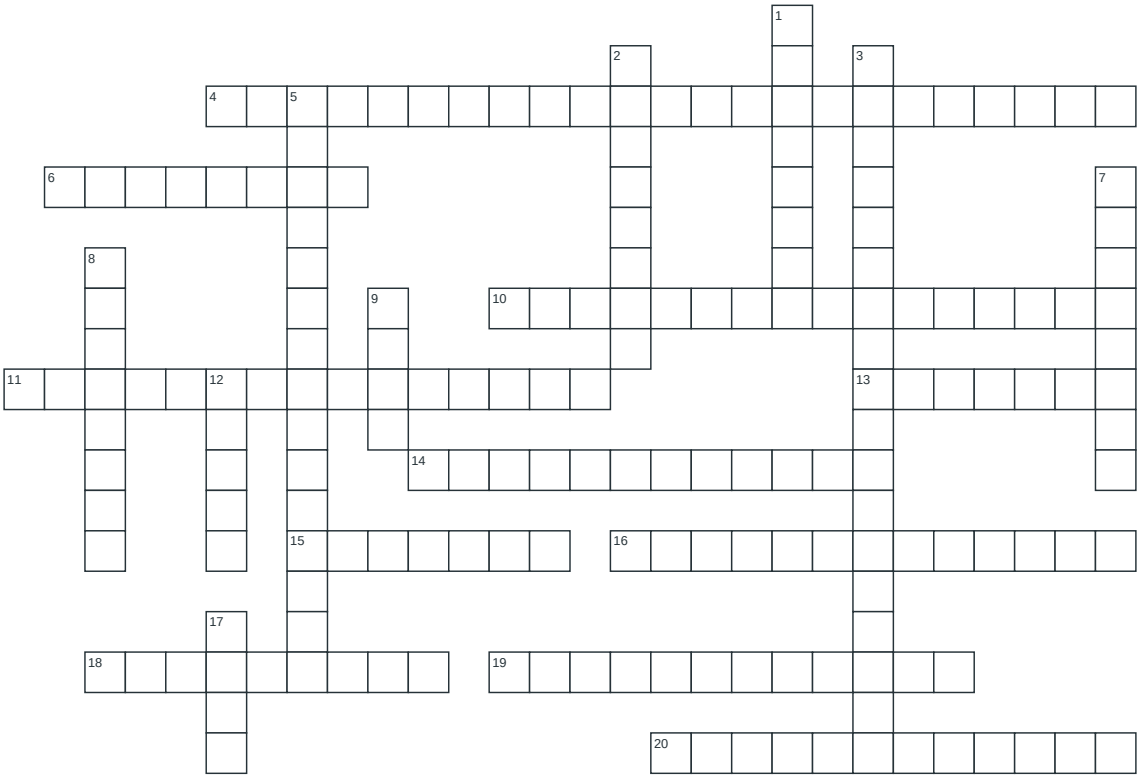
Gravity and Circular Motion Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



ACROSS

4. Becomes one-fourth as large

6. Force that opposes relative motion between surfaces in contact.

10. Inward net force that keeps an object moving along a circular path

11. SI-derived unit commonly used to measure speed or velocity.

13. Resistance to changes in motion

14. Change in velocity per unit time.

15. In the ideal near-Earth projectile model, which force acts after launch when air resistance is neglected

16. Time required for one complete orbit

18. Orientation of motion, needed to describe a velocity vector.

19. The directed change from initial to final position

20. Process by which gravitational force change if one mass doubles and separation stays fixed

DOWN

1. The vector sum of all forces acting on an object

2. Two identical masses move from separation r to separation 3r. What fraction of the original gravitational force remains

3. A description of gravitational force per unit mass at each location

5. A ball's supporting string breaks during horizontal circular motion. Neglect other forces for the instant afterward. Which direction does it initially move

7. Motion in which gravity is the only force acting

8. The total length of a traveled path

9. Measure of the amount of matter in an object

12. Quantity that is a scalar

17. State in which position does not change relative to a chosen reference point.

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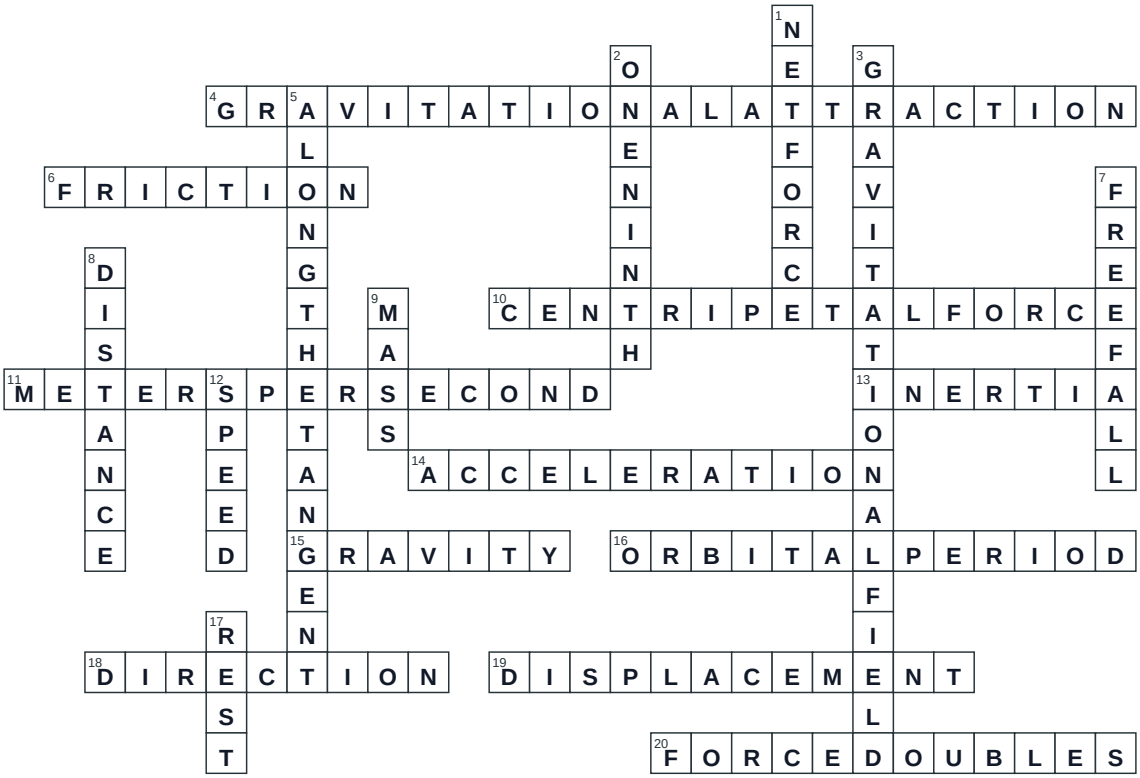
Hard | 20 clues | Teacher Answer Key

TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.



ACROSS

4. Gravitational Attraction
6. Friction
10. Centripetal Force
11. Meters per second
13. Inertia
14. Acceleration
15. Gravity
16. Orbital Period
18. Direction
19. Displacement
20. Force Doubles

DOWN

1. Net Force
2. One-ninth
3. Gravitational Field
5. Along The Tangent
7. Free Fall
8. Distance
9. Mass
12. Speed
17. Rest